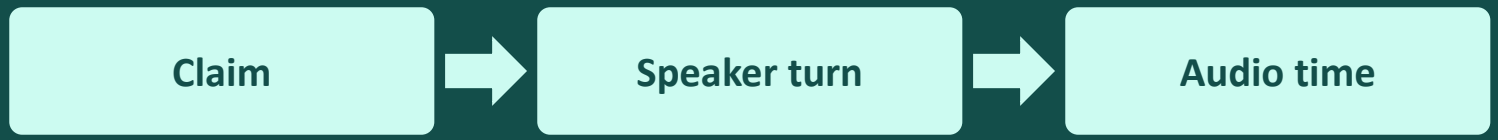
# CallTrace

Evidence-based call review: from a summary claim to the exact turn and audio time

Maharah GDP Trainee Assessment | Call Center Agent Performance Analysis



# A summary is only useful if a reviewer can check it

### 1. Claim in the summary

"The customer accidentally transferred money to the wrong person via the banking app."

### 2. Speaker turn

001-T0002, Customer
"I've realized I've sent it to the wrong place."

### 3. Audio time

Playback starts at 00:18 in the call recording, so the reviewer can listen.

### Required by the assessment

Speaker-separated transcripts, issue + action + outcome summaries, customer sentiment with reasons, for every call

### Added for review

Citations for each claim, sentiment change from beginning to end, and an evaluation that reports failures

Data is role-played, not real customer calls

# Ten sources reviewed; only two passed the eligibility gates

| Source | Result | Deciding reason |
|---|---|---|
| AppTek Call-Center Dialogues | **SELECTED** | Split channels, manual references, CC BY-SA 4.0 |
| SpokenWOZ | **Eligible** | Customer text keeps ASR noise; 8 kHz; CC BY-NC |
| CallCenterEN (AIxBlock) | **Rejected** | No audio released: biometric privacy |
| AxonData samples | **Rejected** | Only 2 open English calls; full set is paid |
| UniData preview | **Rejected** | 5 calls; no derivatives; named firms, user locations |
| InfoBayAI Arabic dual channel | **Rejected** | Gated; about 3 sample calls; terms unread |
| Nakulbh call-center-dataset | **Rejected** | No license or provenance |
| AIxBlock Japanese | **Rejected** | Human-to-machine, not a human agent |
| Arabic call benchmark paper | **Rejected** | No public download located |
| Switchboard / CALLHOME (LDC) | **Rejected** | Licensed; not service calls |

**Weighted score**

**4.90**

AppTek

**3.20**

SpokenWOZ

Author's judgment on 7 stated criteria, out of 5. Ranking holds under equal weights.

Full survey with links, licenses, and 7 papers: reports/research_survey.xlsx

Data is role-played, not real customer calls

# The sample: 14 calls by a fixed rule, and what they cannot show

**14**

calls, one per accent, 14 service domains

**141**

minutes of recordings, drawn from 873 calls

**19,248**

manually transcribed reference words

**0**

calls excluded or replaced after sampling

Rule: seed 20261108; shuffle each accent; first call in an unused domain. Metadata only, so calls cannot be chosen for being easy.

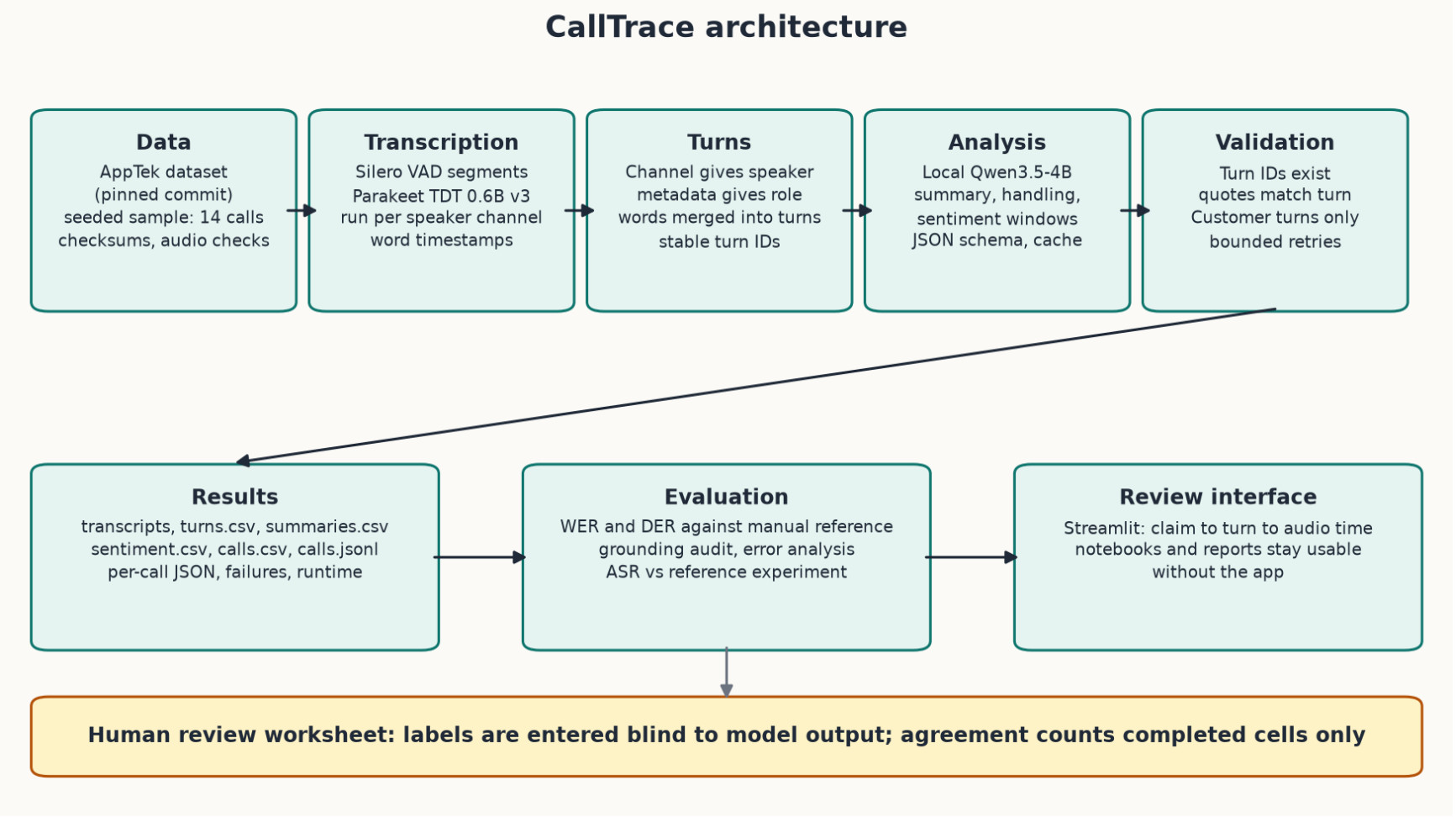
## Role-played, not real customer calls

- No claims about real customers or business impact
- Sentiment describes language in role-plays
- Clean split channels: mono phone calls would be harder
- 14 calls cannot support per-accent claims

### Data checks that changed the design

- Channel 1 is the customer in 66 of 873 calls: order is not role
- Call 005: channels appear out of sync (54% overlap in the manual reference)

# One readable pipeline, with no paid services




Data is role-played, not real customer calls

# Speaker, role, and turn boundaries each have a stated source

| Item | Source | Note |
|---|---|---|
| Speaker 1 / 2 | Dataset split channels | One speaker per file |
| Customer / Agent | Channel metadata | Agent-phrase check agrees: 14 of 14 |
| Turn boundaries | Predicted from ASR words | Stable IDs and timestamps |
| Reference text | Manual, kept apart | Used only for evaluation |

## Speaker error against manual segments (14 calls)

| Method | Median DER | Median confusion | Max confusion |
|---|---|---|---|
| Split channels (used) | 10.23% | 0.14% | 1.1% |
| Mono baseline (contrast) | 6.97% | 1.1% | 35.23% |

Baseline clusters needed the reference for mapping. Channel DER is mostly false alarm, as word ends run long.

### Call 001, generated transcript

**001-T0001 00:13 Agent (Speaker 1)**

Hi, thanks for calling A and Z. You are speaking with Tom. How can I help you today?

**001-T0002 00:18 Customer (Speaker 2)**

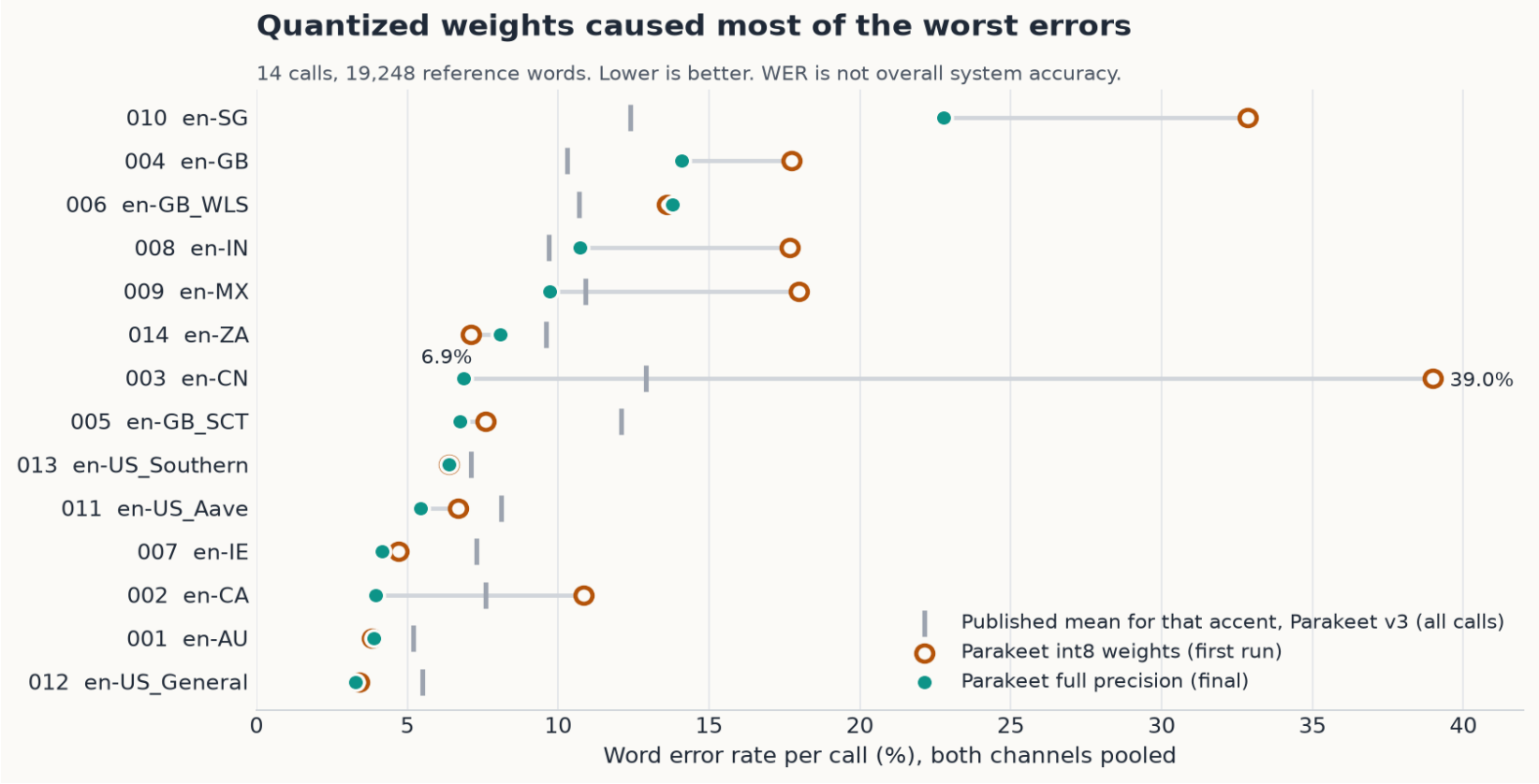
Hey Tom, uh my name's Laura. I've just been on the banking app and I've made a mess of things to be honest.

**001-T0004 00:51 Agent (Speaker 1)**

Okay, um I can understand how frustrating that's been.

Data is role-played, not real customer calls

# A benchmark caught a mistake that a pilot check had missed




**14.39%**

pooled WER, int8 weights

**9.32%**

pooled WER, full precision (19,248 words)

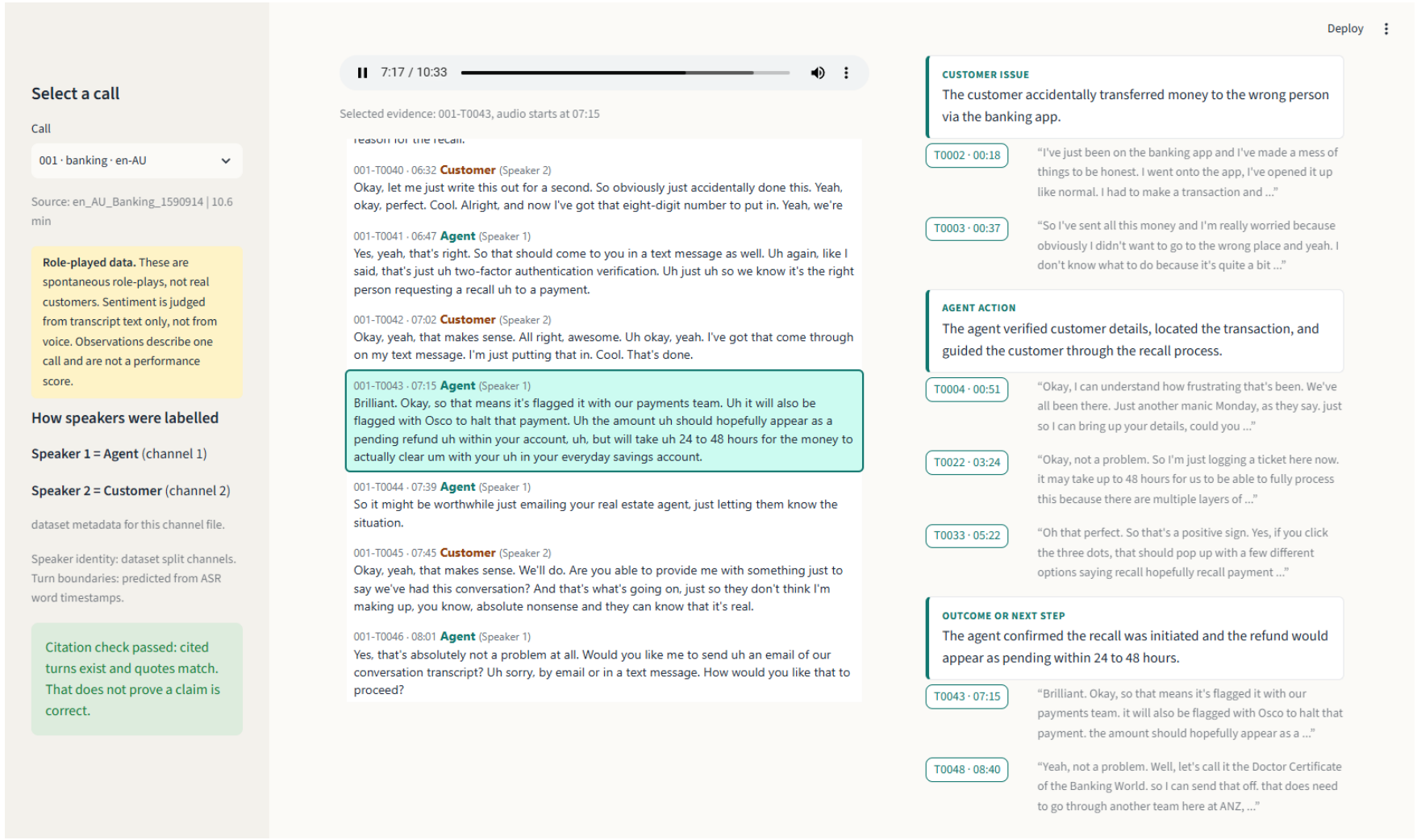
### How it was found

- VAD or level: no effect
- Whisper: 6.83% vs 45.78%
- Full precision: 5.38%

Published average 9.2% covers all 873 calls, so this is a sanity check.

Data is role-played, not real customer calls

# Live demo: follow a claim to the turn and the audio




## In the app

1. Select a call
2. Read issue, action, outcome
3. Click a citation
4. Turn highlights; audio seeks
5. Inspect sentiment windows
6. Read limitations

Real capture from the running app: citation T0043 selected, audio playing from 07:15.

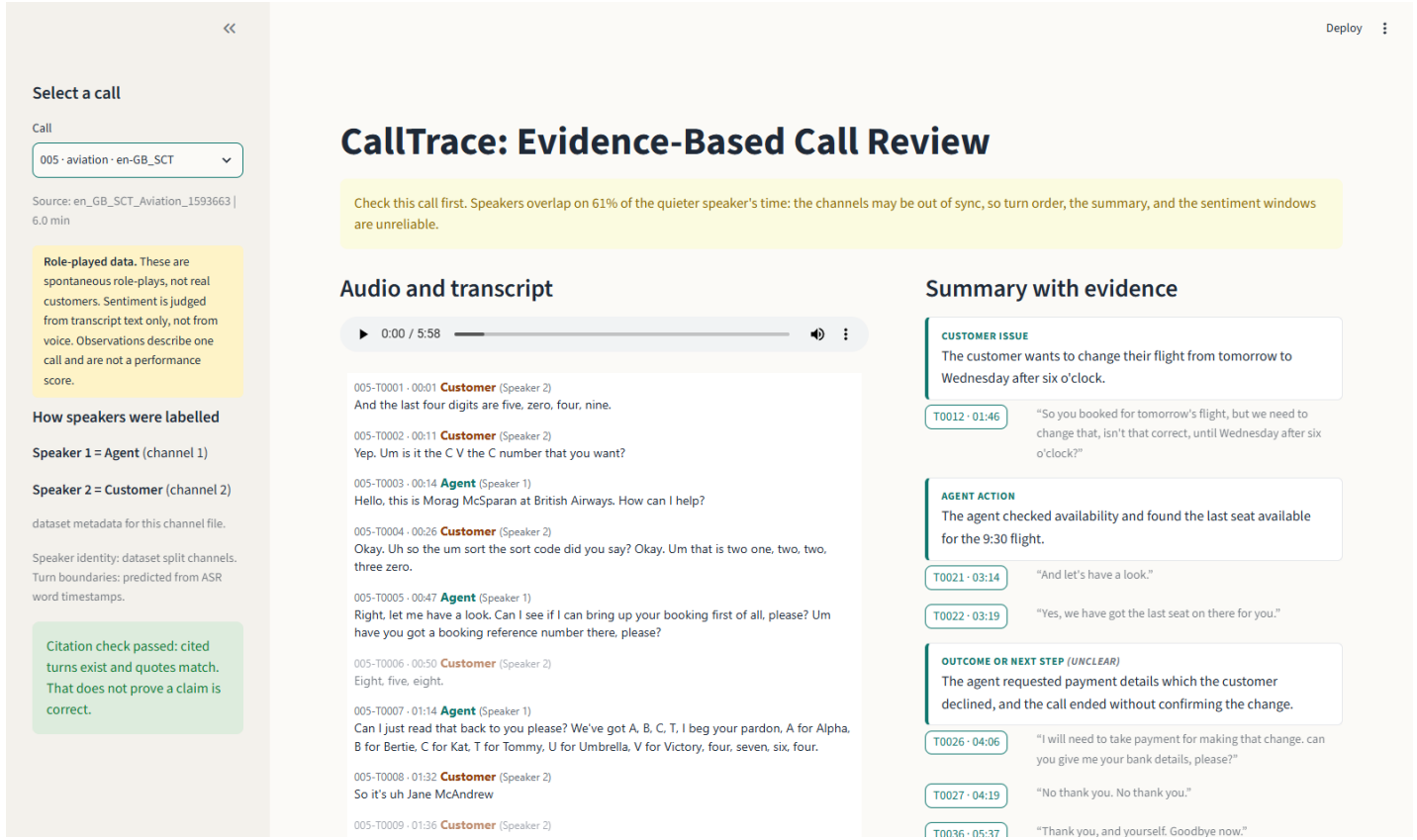
# Citations exist, but only 20 of 42 claims are fully covered

| Check | Result | Sample | Compared against |
|---|---|---|---|
| Pooled WER (agent / customer) | 9.32% (5.79% / 15.04%) | 19,248 words, 14 calls | Manual reference |
| Speaker confusion, median / max | 0.14% / 1.1% | 14 calls, 0.25 s collar | Manual segments |
| Calls passing citation checks | 11 of 14 | 275 citations | Automatic validator |
| Claims fully supported by cited turns | 20 of 42 | partial 22, unsupported 0 | **AI audit, not human** |
| Claims correct against the whole call | 34 of 42 | partly 7, wrong 1 | **AI audit, not human** |
| Sentiment: overall / beginning / ending | 8 / 13 / 12 of 14 | Reader saw model labels first | **AI reader, not human** |
| Sentiment against human labels | Not available | 0 labels completed | **Worksheet pending** |
| Labels unchanged, ASR vs reference | 20 of 25 | 5 calls, 5 labels each | Same analysis settings |

Runtime on this laptop (Ryzen 7 6800H, RTX 3050 Ti 4 GB): about 7.5 s of ASR per audio minute on CPU; about 46 s of analysis per call. ASR model load: about 5 s.

Data is role-played, not real customer calls

# A valid citation can still support a wrong conclusion




Real screenshot, call 005: the flag sits above a citation check that still passes.

### Sync problem in the data

The reference has the same order. Only a reader can see the conclusion is wrong.

### Sentiment leans on politeness

Overall is positive in 10 of 14 calls; AI reader agrees on only 8 of 14.

### Labels can flip

Call 012 (3.28% WER) changes from positive to neutral on the reference.

### ASR errors can reach summaries

“Prescription review” in call 003 was really “refill”.

Data is role-played, not real customer calls

# Reproducible today; human review next

## Next steps

- Human labels from the blind worksheet
- Real, consented calls
- Mono audio with predicted speakers

## Run it

```
python scripts/start_llm_server.py
python -m calltrace run --smoke
python -m calltrace run --full
streamlit run app/streamlit_app.py
```

36 tests pass | 4 notebooks execute in order | pinned models, data commit, and seed